

DE Lab10
Hayden Fuller

Q1:

a)

i)

$$V_{B4} = R_7 \cdot I_{R7} = R_7 \cdot ((V_{EE} + 2V_D)/(R_7 + R_8)) = 1k \cdot (-3.8/6k) = -3.8/6 = -0.6333V$$

$$V_R = V_{B4} - V_{BE} = -0.6333 - 0.7 = -1.3333V$$

$$V_{IL} = V_R - .35 = -1.6833V$$

$$V_{IH} = V_R + .35 = -0.9833V$$

when A=B=low:

$$V_X = V_R - V_{BE} = -1.3333 - 0.7 = -2.0333V$$

$$I_{R3} = (V_X - V_{EE})/R_3 = (-2.0333 - -5.2)/850 = 3.72549mA$$

$$V_{O2H} = 0$$

$$V_{O2L} = 0 - I_{R3} \cdot R_2 = -0.7824V$$

when A||B=high

$$V_X = V_{IH} - V_{BE} = -0.9833 - 0.7 = -1.6833V$$

$$I_{R3} = (V_X - V_{EE})/R_3 = (-1.6833 - -5.2)/850 = 4.13726mA$$

$$V_{B5} = 0 - I_{R3} \cdot R_1 = -0.78608V$$

$$V_{O1H} = 0 - V_{BE} = -0.7V$$

$$V_{O1L} = V_{B5} - V_{BE} = -1.48608V$$

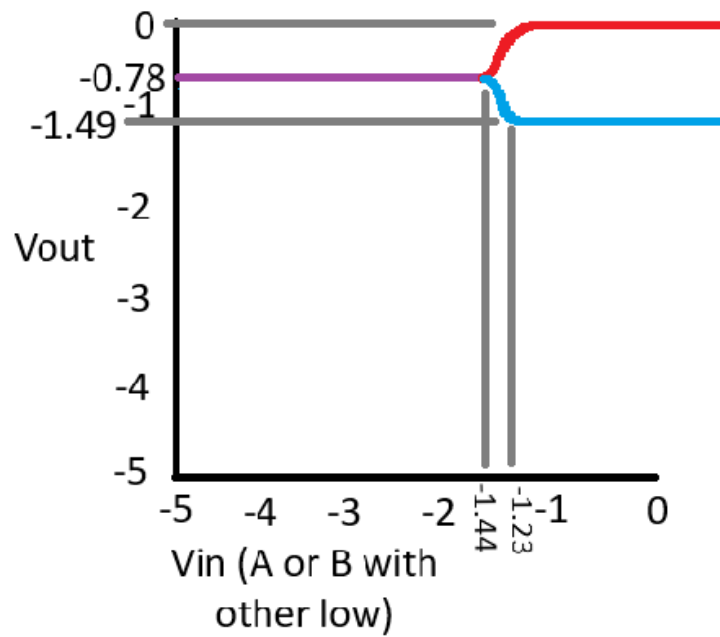
ii)

$$V_{swing} = I_{R3(00)} \cdot R_1 + I_{R3(A||B)} \cdot R_2 = 3.72549m \cdot 190 + 4.13726m \cdot 210 = 1.57666V$$

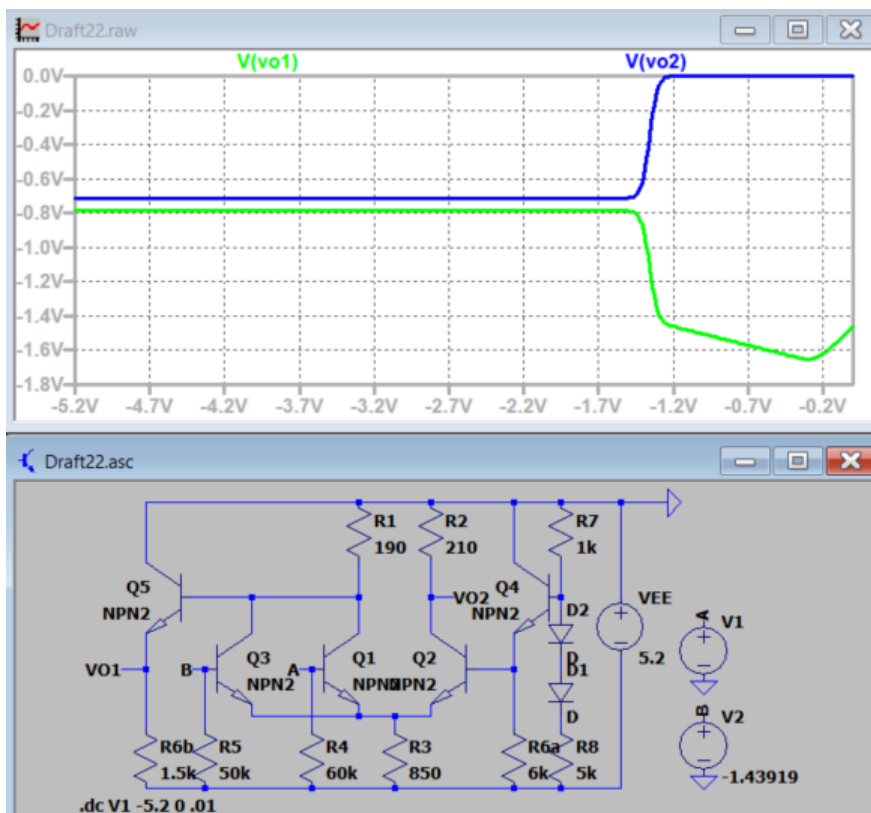
$$V_{IH} = V_R + V_T \cdot \ln(V_{swing}/V_T - 2) = -1.22748V$$

$$V_{IL} = V_R - V_T \cdot \ln(V_{swing}/V_T - 2) = -1.43919V$$

b)
i)



ii)



the plots are similar, VO1 drops between VIL and VIH

c)

$$NMH = V_{OH} - V_{IH} = -0.7 - (-1.22748) = 0.52748V$$

$$NML = V_{IL} - V_{OL} = -1.43919 - (-1.48608) = 0.04689V$$

d)

$$VO1 = A \text{ NOR } B$$

$$VO2 = A \text{ OR } B$$

e)

i)

$$NMH' = NMH/2 = 0.52748/2 = 0.26374V$$

$$VOH' = NMH' + V_{IH} = 0.26374 + (-1.22748) = -0.96374V$$

use $V_{in} = VOH' = -0.96374V$:

$$V_X = VOH' - V_{BE} = -0.96374 - 0.7 = -1.66374V$$

$$I_{R3} = (V_X - V_{EE})/R_3 = (-1.66374 - (-5.2))/850 = 4.16031mA$$

$$I_{B1} = I_{E1}/(B+1) = I_{R3}/(B+1) = 4.16031mA/81 = 51.3618\mu A$$

$$I_{R4} = (-0.96374 - (-5.2))/60k = 70.6043\mu A$$

$$I_{IHA} = I_{B1} + I_{R4} = 121.966\mu A$$

$$I_{R5} = (-0.96374 - (-5.2))/50k = 84.7252\mu A$$

$$I_{IHB} = I_{B1} + I_{R5} = 136.087\mu A$$

$$I_{OH} = I_{E5} - I_{R6}$$

$$I_{E5} = I_{B5} \cdot (B+1) = (B+1) \cdot -(I_{OH}' + V_{BE})/R_1 = 81 \cdot (0.96374 - 0.7)/190 = 112.4365mA$$

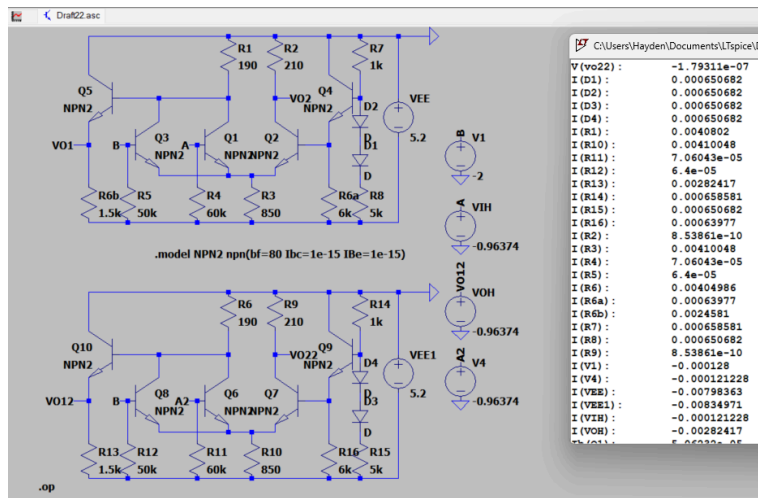
$$I_{R6} = -0.96374 - (-5.2) = 2.82417mA$$

$$I_{OH} = 112.4365mA - 2.82417mA = 109.612mA$$

$$N_A = I_{OH}/I_{IHA} = 898$$

$$N_B = I_{OH}/I_{IHB} = 805$$

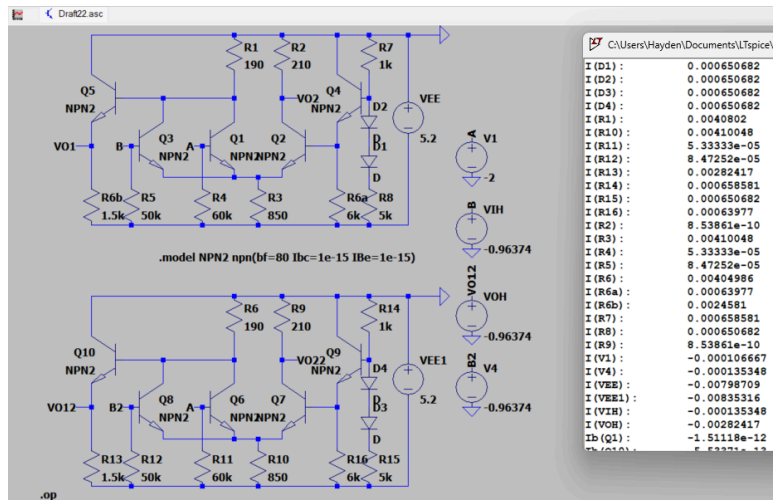
ii)



$$I_{IHA} = -0.000121228$$

$$I_{OHA} = -0.00282417$$

$$N_A = I_{OHA} / I_{IHA} = 23.30$$



$$I_{IHB} = -0.000135348$$

$$I_{OHB} = -0.00282417$$

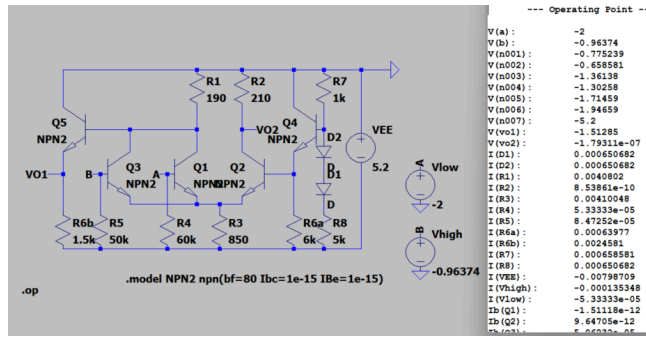
$$N_B = I_{OHB} / I_{IHB} = 20.86$$

I_{IH} calculated at 122uA and 136uA, matches

I_{OH} much lower than calculated due to output saturation current

N lower than calculated, but still higher with A input.

f)



$$P = I(V_{EE}) * V_{EE} + I(A) * V(A) + I(B) * V(B) = 7.98709\text{m} * 5.2 + 53.3333\text{u} * 2 + 135.348\text{u} * 0.96374$$

$$P = 41.533\text{m} + 106\text{u} + 130\text{u} = 41.769\text{mW}$$

2)

SA	V1	!V1	V2	!V2
00	1	1	0	1
01	1	1	1	0
10	0	1	1	1
11	1	0	1	1

demux